

WTW158 2017 Worksheet on vector algebra

1. Given $\vec{a} = \langle 2, 2, 0 \rangle$, $\vec{b} = \langle -1, 1, 0 \rangle$ and $\vec{c} = \langle 0, 1, 1 \rangle$.
 - 1.1 Explain why $\vec{a}$ and $\vec{b}$ form two sides of a right-angled triangle.
 - 1.2 Find the angle between $\vec{b}$ and $\vec{c}$.
2. Given the vectors $\vec{a} = \langle 2, -1, 0 \rangle$ and $\vec{b} = \langle 0, 1, 2 \rangle$ find
 - 2.1 $\vec{a} \times \vec{b}$, $\vec{b} \times \vec{a}$ and $\vec{b} \times \vec{b}$
 - 2.2 $\vec{a} \cdot (\vec{a} \times \vec{b})$
3. Given the points $P(3, 1, 4)$ and $Q(-2, -2, 3)$.
 - 3.1 Find the distance between the point P and the xy -plane.
 - 3.2 Find the distance between the point P and the x -axis.
 - 3.3 Find the vector $\overrightarrow{QP}$.
4. Given the vectors $\vec{a} = \langle 1, 1, 4 \rangle$ and $\vec{b} = \langle 0, 2, -3 \rangle$
 - 4.1 Find a unit vector in the direction of $\vec{a}$.
 - 4.2 Find a vector of length 5 in the opposite direction to $\vec{a}$.
5. Find the angle between the vectors $\vec{a} = \langle 0, 1, -1 \rangle$ and $\vec{b} = \langle -1, -1, 0 \rangle$.
6. The angle between two unit vectors $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$. Find $\vec{a} \cdot \vec{b}$.
7. Apply the right hand rule to find:
 - 7.1 $\vec{i} \times \vec{j}$ 7.2 $\vec{k} \times \vec{i}$
8. Find s such that $\vec{a} = \langle 0, s, -1 \rangle$ and $\vec{b} = \langle -1, s, 1 \rangle$ are orthogonal.
9. Find the area of the parallelogram formed by $\vec{a} = \langle 0, 1, -1 \rangle$ and $\vec{b} = \langle -1, -1, 0 \rangle$ as two of the sides.
10. Say whether each of the following is true or not and explain why.
 - 10.1 $(\vec{a} \times \vec{b}) \cdot \vec{b} = 0$ for all vectors $\vec{a}$ and $\vec{b}$.
 - 10.2 $(\vec{u} + \vec{v}) \times \vec{v} = (\vec{u} + \vec{v}) \times \vec{u}$ for all vectors $\vec{v}$ and $\vec{u}$

Answers

- 1.1 $\vec{a} \cdot \vec{b} = 0$
- 1.2 $\theta = \frac{\pi}{3}$
- 2.1 $\langle -2, -4, 2 \rangle$, $\langle 2, 4, -2 \rangle$, $\langle 0, 0, 0 \rangle$
- 2.2 0
- 3.1 4
- 3.2 $\sqrt{17}$
- 3.3 $\langle 5, 3, 1 \rangle$
- 4.1 $\left\langle \frac{1}{\sqrt{18}}, \frac{1}{\sqrt{18}}, \frac{4}{\sqrt{18}} \right\rangle$
- 4.2 $\left\langle -\frac{5}{\sqrt{18}}, -\frac{5}{\sqrt{18}}, -\frac{20}{\sqrt{18}} \right\rangle$
5. $\frac{2\pi}{3}$
6. $\frac{1}{\sqrt{2}}$
- 7.1. $\vec{k}$ 7.2 $\vec{j}$
8. $s = \pm 1$
9. $\sqrt{3}$
- 10.1 Yes, $\vec{a} \times \vec{b} \perp \vec{b}$
- 10.2 No, LHS and RHS have opposite signs